Seonghan Kim

Website | GitHub

Location: Seoul, Korea
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Summary

My research focuses on machine learning for computer vision, the bio domain, and on-device inference. In my master's program, I aim to study model compression and optimization, how training data affects model performance, and reinforcement learning.

Education

B.S. in Electronic Engineering (1st major) and Human-Centered AI Engineering (2nd major) — Seoul, Korea
Sangmyung University — *Expected Feb 2027*

Publications (* equal contribution)

Manuscripts Under Review

1. *Dissecting Tissue-Context Effects in Graph Neural Networks for Rat K_p Prediction: Additive Shifts, Fusion Position, and Extrapolation*
   Seonghan Kim*, Hyunseo Cho*, Hyeoncheol Jeong*, Moonkyu Jeong, Kyongcheol Ko, Kyeongryoon Lee, Daesik Jeong
   Journal of Pharmacokinetics and Pharmacodynamics (**JPKPD**) [SCIE, Q2]
   Under review
2. *The choice of species representation in multi-species aquatic acute-toxicity prediction is constrained by data availability rather than by predictive accuracy*
   Seonghan Kim, Kyungmin Jeon, Hyunseo Cho, Seongyun Park, Daesik Jeong
   Journal of Cheminformatics (**JCheminf**) [SCIE, Q1] [preprint]
   Under review

Conference Publications

1. *Implementation and Processing-Time Analysis of an NPU-Based Real-Time Iris Recognition System*
   Huijae Bae*, Eunsu Yun*, Dongwon Lee, SeonKyun Park, **SeongHan Kim**, JongWeon Kim, DaeSik Jung
   Proceedings of the Institute of Electronics and Information Engineers (**IEIE**) Summer Conference, pp. 2201–2205, 2025 (in Korean) [dbpia]

Research Experience

Undergraduate Researcher — *Mar 2025 – Dec 2026*
Dept. of Computer Science, Sangmyung University

Intern — *Jul 2025 – Aug 2025*
InSiliCox

- Led frontend development of OMNIDRUG, a drug-discovery platform; assisted with docking and molecular dynamics while learning them.

Undergraduate Researcher — *Jun 2024 – Dec 2024*
OpenLab, Dept. of Emotion Engineering, Sangmyung University

Research Projects

Skin Image Labeling Tool — *Oct 2026 – Dec 2026*
Lameditech — Industry project

- Labeling program for building skin-condition (e.g., wrinkle) datasets: face detection and mesh-map generation with MediaPipe, with ROI-cropping rules designed on top of the mesh.

Mouse Filler Volume Change Measurement & Prediction — Lead Researcher — *May 2026 – Jun 2026*
Gachon University, College of Korean Medicine — Industry project

- **Role:** Owned all Instant-NGP work and the extraction of filler-injection regions (volume measurement excluded).
- Reconstructed 3D scenes from 2D videos with Instant-NGP and extracted the filler-injection regions.

Aging & Skin-Trouble Progression Prediction App — Lead Researcher — *Apr 2026 – Jun 2026*
Amway Global — Industry project

- **Role:** Made the key decisions across planning and development, and served as the primary point of contact with Amway.
- **Outcome:** Journal paper on aging prediction in preparation

Bifusion *Mar 2026 – Nov 2026*

Capstone Project [github] Team project

- **Role:** Team lead; concept design; owned all AI components including the augmentation and few-shot models.
- Designed a Normal/Anomaly classifier trainer that fine-tunes the denoising U-Net of the medical diffusion model Medfusion on user data and combines the generated images with the few-shot model ProtoNet.

Cammand *Mar 2026 – Sep 2026*

Capstone Project Team project

- **Role:** Planned the overall project and designed the gesture computation algorithm.
- MediaPipe-based gesture classification combined with an arithmetic gesture-recognition method.

NPU Algorithm for Toxicity Prediction — Lead Researcher *Oct 2025 – Dec 2025*

Korea Research Institute of Bioscience and Biotechnology (KRIBB) Industry project

- **Role:** Led every stage except data-related work.
- Selected the operators best suited to the Kneron KL630 NPU and built an MLP from those operators alone to predict molecular toxicity.
- **Outcome:** Follow-up first-author manuscript on aquatic toxicity (J. Cheminformatics, under review)

Tissue-specific Kp Prediction Model — Lead Researcher *May 2025 – Oct 2025*

Korea Research Institute of Bioscience and Biotechnology (KRIBB) Industry project

- **Role:** Owned model and experiment design, development, and analysis on KRIBB data; co-authors handled data preprocessing and collection, and paper figures and tables.
- Developed a Chemprop-based GNN for predicting tissue-to-plasma partition coefficients (Kp) and built a Streamlit demo page.
- **Outcome:** Co-first-author manuscript under review at JPKPD

DoorBox — Crime-prevention Smart Doorbell *Mar 2025 – Jul 2025*

Team Project Team project

- **Role:** Team lead and main programmer; quantization and NPU optimization.
- Built a face-analysis pipeline: YOLOv5s face detection quantized with the Kneron SDK (Docker) and deployed on the KL630 NPU for real-time detection and coordinate transmission; emotion (alert / non-alert, EfficientNet-B0) and gender and age-group classifiers (MobileNetV3-Small distilled from RegNetY-1.6GF) ran on the CPU.
- **Results:** Face detection mAP@0.5 0.861 (YOLOv5s fine-tuned on CelebA); validation accuracy of 98.42% for gender and 69.10% for age group with lightweight student models trained by knowledge distillation.

NPU-based Real-time Iris Recognition System *May 2024 – Aug 2024*

Hyundai MIB International Industry project

- Improved iris segmentation by applying a Gaussian filter for eyelash removal, and ported the Python pipeline to C.
- **Results:** Segmentation IoU 0.94–0.97 and mAP50 0.99 for iris, pupil, and eyelid; the segmentation stage (incl. pre/post-processing) took 47.4 ms per image on the NPU (about 21 FPS), roughly 7× faster than Jetson Nano with TensorRT (330.7 ms).
- **Outcome:** IEIE Summer Conference paper (co-author)

Research Grants (participant)

Global CORE-AX: Securing Core Copyright Technologies and Fostering Convergence Talent through Joint Research *Aug 2026 – Dec 2026*

Ministry of Culture, Sports and Tourism — Copyright Workforce Development Program

Excellent Young Researcher (Seed Research) grant *Sep 2025 – Aug 2026*

National Research Foundation of Korea (NRF)

Honors & Awards

- Human AI Studio Festival 2026 — Top Excellence Award (Graduation Exhibition)

Activities

- Student Council President, Intelligent IoT Convergence / Electronic Engineering (2025)
- Student Council Vice President (2024.09 – 2024.12)
- Hackathon Instructor, Sangmyung Univ. Affiliated Girls' High School (2024.08)
- INIRO — Programming Club (2021 – 2025)

Additional Information

- **Languages:** TOEIC Speaking IH (150), valid until 2028.07
- **Military Service:** Republic of Korea Army, 3rd Infantry Division — Sergeant, honorably discharged